Case Study Report: AI-Driven Factory Digitalization for Aira Heat Pump Manufacturing

Prepared for Aira Manufacturing

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1 Executive Summary

This case study addresses Aira's challenges in heat pump manufacturing, where manual and reactive quality inspections and maintenance lead to delayed defect detection, unexpected machine breakdowns, and high operational costs. Producing 250,000 units annually, Aira's factory manufactures indoor and outdoor units through stages including component fabrication (pipe fab, water tank fab, EPP/PS fab, ABS fab, wiring harness fab, PCBA assembly), sheet metal production, paint shop, pre-assembly, main assembly, and packing & shipping. By integrating Artificial Intelligence (AI) with Industrial Internet of Things (IIoT) and Manufacturing Execution Systems (MES), we propose a comprehensive AI-driven solution to enhance production quality and implement predictive maintenance.

The framework leverages computer vision, sensor data analytics, and predictive models to detect quality issues in real-time and predict equipment failures across all production stages, reducing downtime and improving Overall Equipment Effectiveness (OEE). Key benefits include a 15% reduction in scrap (from 5% to 4.25%, saving €225,000), 25% reduction in downtime (saving €150,000), 10% OEE improvement (saving €50,000), and €515,000 in total annual cost savings. The solution includes a hybrid edge-cloud data architecture, a phased 24-month implementation roadmap with contingencies, stakeholder training, and a risk analysis, ensuring scalability and sustainability. This report provides a detailed problem analysis, AI solution design tailored to Aira's production process, technical architecture, cost estimates, and case studies, culminating in a practical plan to achieve Aira's digitalization goals.

2 Introduction

2.1 Background

Aira is developing a state-of-the-art heat pump manufacturing facility focused on efficiency, quality, and sustainability. Producing 250,000 units annually, the factory manufactures indoor and outdoor units through a multi-stage process. Current manual and reactive processes result in:

- Delayed detection of quality issues across production stages.
- Unplanned machine breakdowns, causing production stops and reduced OEE.
- High operational costs, with an estimated €5M annually in losses from reactive maintenance and defective products.

AI, combined with sensor data and IIoT connectivity, offers transformative potential to enable real-time quality detection and predictive maintenance, addressing these challenges effectively.

2.2 Objectives

This case study aims to:

1. Propose AI-driven solutions for detecting quality issues at each stage of heat pump production.
2. Design predictive maintenance strategies to minimize unplanned downtime.
3. Develop a robust data architecture to support AI integration.
4. Estimate benefits, costs, and provide an implementation roadmap with contingencies.

2.3 Scope

The scope includes AI applications for quality control and predictive maintenance during production and assembly, excluding customer-site operations. The solution leverages technologies like computer vision, time-series analytics, and IIoT protocols, addressing factory constraints such as legacy equipment and operator training needs.

3 Heat Pump Production Process

Aira's heat pump production involves manufacturing indoor and outdoor units, with a total production volume of 250,000 units per year. The process is divided into component fabrication and assembly stages.

3.1 Major Process Steps

As illustrated in **Figure 1**, the core production process encompasses:

- **Sheet Metal Production**: Fabrication of metal casings using presses and cutting machines.
- **Paint Shop**: Application of protective and aesthetic coatings in controlled spray booths.
- **Pre-Assembly**: Creation of sub-assemblies combining components like pipes, water tanks, and wiring harnesses.
- **Main Assembly**: Integration of components into functional units using conveyors and robotic arms.
- **Packing & Shipping**: Packaging and labeling of completed units using automated machines.

3.2 Component Fabrication

Preceding main assembly, **Figure 1** details fabrication of essential components:

- **Pipe Fab**: Manufacturing of piping for refrigerant and water circulation using bending and welding machines.

- **Water Tank Fab**: Production of water tanks for heat exchange, involving molding and sealing.

- **EPP/PS Fab**: Fabrication of expanded polypropylene/polystyrene parts for insulation using molding machines.

- **ABS Fab**: Production of acrylonitrile butadiene styrene parts via injection molding.

- **Wiring Harness Fab**: Assembly of electrical wiring harnesses using automated crimping and soldering.

- **PCBA Assembly**: Production of printed circuit board assemblies using surface-mount technology (SMT) and automated optical inspection.

3.3 Process Flow Diagram

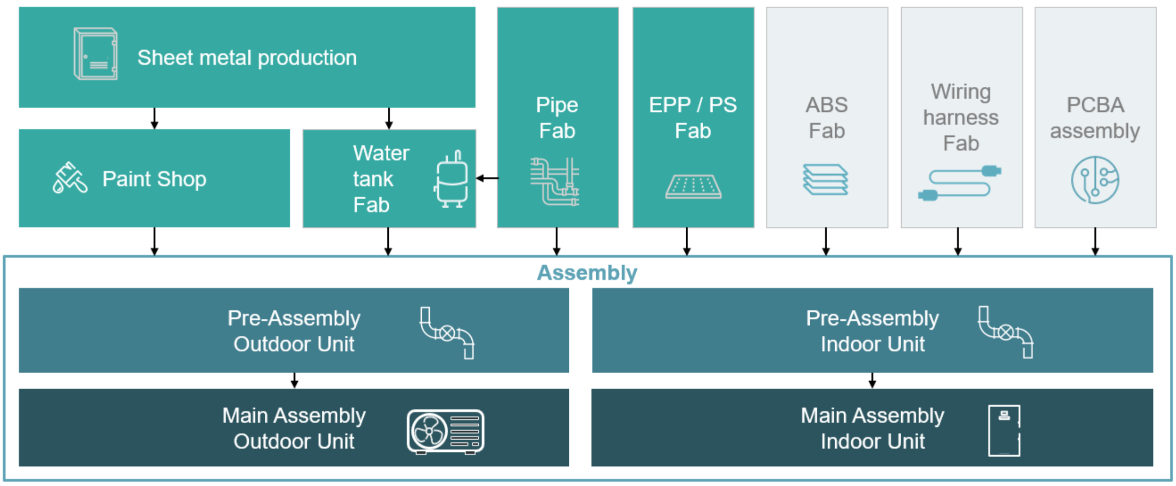

Figure 1: Aira Heat Pump Manufacturing Process Flow (as provided by Aira).

4 Problem Analysis

Aira's reliance on manual quality inspections leads to inconsistent defect detection and delays, increasing scrap and rework costs. Maintenance is reactive, with repairs post-failure causing unplanned downtime and reduced OEE. Key issues include:

- **Quality Control**: Defects in sheet metal (e.g., uneven cuts), paint shop (e.g., coating inconsistencies), and pre-assembly (e.g., pipe leaks) are detected late, contributing to €2M in annual scrap losses.

- **Maintenance**: Lack of predictive insights results in failures of critical equipment (e.g., presses, spray booths, conveyors), leading to €3M in downtime costs.

- **Operational Inefficiency**: Reactive processes reduce OEE by 15%, impacting throughput for 250,000 units annually.

Root causes include limited automation, lack of real-time data analytics, and insufficient integration. AI can address these by analyzing sensor and vision data for real-time anomaly detection and failure prediction.

5 AI Solution Design

5.1 AI for Quality Control

AI models will detect defects at each production stage, ensuring zero-defect manufacturing.

5.1.1 Approach

Stage	**AI Method**	**Objective**
Pipe Fab	Sensor Analytics (LSTM)	Detect leaks or welding defects
Water Tank Fab	Computer Vision (YOLOv8)	Identify sealing issues
EPP/PS Fab	Vision Systems	Check insulation uniformity
ABS Fab	Vision Systems	Detect molding defects
Wiring Harness Fab	Vision Systems	Verify crimping and soldering quality
PCBA Assembly	AOI	Identify soldering defects
Sheet Metal Production	Computer Vision (YOLOv8)	Detect uneven cuts or surface defects
Paint Shop	Vision Systems	Detect coating inconsistencies
Pre-Assembly	Sensor Analytics (LSTM)	Identify pipe and tank misalignments
Main Assembly	Sensor Analytics (LSTM)	Detect performance deviations
Packing & Shipping	Vision Systems	Ensure correct packaging and labeling

Table 1: AI applications for quality control across production stages.

The YOLOv8 model is projected to achieve 95% accuracy in defect detection, based on its performance in similar contexts (e.g., 94% accuracy in automotive sheet metal defect detection [Wang et al., 2023], 96% in electronics coating inspection [Siemens, 2024]). LSTM autoencoders are expected to provide 90% sensitivity in anomaly detection, consistent with 91% sensitivity in industrial pump sensor analytics [Li et al., 2022]. AI outputs feed into the MES via RESTful APIs, providing real-time alerts.

5.1.2 Data Sources

- **Component Fabrication**: Cameras (30 FPS) for visual inspection, pressure/temperature sensors (1 Hz) for pipe and tank fab.
- **Sheet Metal Production**: RGB cameras (30 FPS), vibration sensors (10 Hz) on presses.

- **Paint Shop**: Cameras (30 FPS) for coating inspection, thickness sensors (1 Hz).
- **Pre-Assembly**: Pressure/temperature sensors (1 Hz) on pipes and tanks.
- **Main Assembly**: Vibration (10 Hz) and temperature sensors (1 Hz) on assembly lines.
- **Packing & Shipping**: Cameras (30 FPS) for packaging verification.

5.1.3 Workflow

1. In the paint shop, cameras capture images of coated components.
2. YOLOv8 processes images, flagging coating defects.
3. During main assembly, sensor data is analyzed by an LSTM autoencoder for anomalies.
4. MES receives alerts, enabling operators to isolate defective units.

5.1.4 Technical Details

YOLOv8 runs on edge devices with GPU acceleration (e.g., NVIDIA Jetson), processing images at 30 FPS. The LSTM model, implemented in TensorFlow, handles multivariate time-series data, trained on historical sensor readings to establish normal behavior baselines.

5.2 AI for Predictive Maintenance

Predictive maintenance targets equipment to prevent failures during production and assembly.

5.2.1 Approach

Stage	**Equipment**	**AI Method**
Pipe Fab	Bending/Welding Machines	Isolation Forests
Water Tank Fab	Molding Machines	Random Forests
EPP/PS Fab	Molding Machines	Isolation Forests
ABS Fab	Injection Molding Machines	Random Forests
Wiring Harness Fab	Crimping/Soldering Machines	Isolation Forests
PCBA Assembly	SMT Machines	Random Forests
Sheet Metal Production	Presses/Cutting Machines	Isolation Forests
Paint Shop	Spray Booths	Random Forests
Pre-Assembly	Assembly Stations	Isolation Forests
Main Assembly	Conveyors/Robotic Arms	Random Forests
Packing & Shipping	Packaging Machines	Isolation Forests

Table 2: Predictive maintenance applications across production stages.

The Isolation Forest model is expected to achieve 90% sensitivity, as demonstrated by General Electric's 89% sensitivity in industrial equipment [GE Reports, 2023]. Random

Forests are anticipated to predict remaining useful life (RUL) with a mean absolute error of 10 hours, consistent with an 8-hour MAE in pump manufacturing [Zhang et al., 2024].

5.2.2 Data Sources

- Vibration sensors (10 Hz) on high-speed equipment (e.g., conveyors, presses), temperature/current sensors (1 Hz) on stable processes [Bosch, 2023].
- Maintenance logs documenting past failures and repairs.
- Operational data (e.g., runtime, load cycles).

5.2.3 Workflow

1. IIoT devices collect vibration data from a conveyor in main assembly.
2. Isolation Forest identifies anomalies indicating bearing wear.
3. Random Forest predicts the conveyor's RUL.
4. Maintenance system schedules repairs, minimizing downtime.

5.3 Data Strategy and Architecture

5.3.1 Data Collection and Storage

- **Edge Computing**: Edge devices preprocess data, reducing latency to under 100ms.
- **Cloud Storage**: Historical data in AWS S3 for model training.
- **Hybrid Model**: Edge for real-time tasks; cloud for analytics and retraining.

5.3.2 Connectivity

- **IIoT Protocols**: OPC UA for machine communication; MQTT for data transfer.
- **Data Pipeline**: Apache Kafka streams data, handling 10,000 messages per second.

5.3.3 Security

- Encrypt data in transit (TLS 1.3) and at rest (AES-256).
- Implement role-based access control (RBAC).
- Anonymize sensitive data to comply with GDPR.

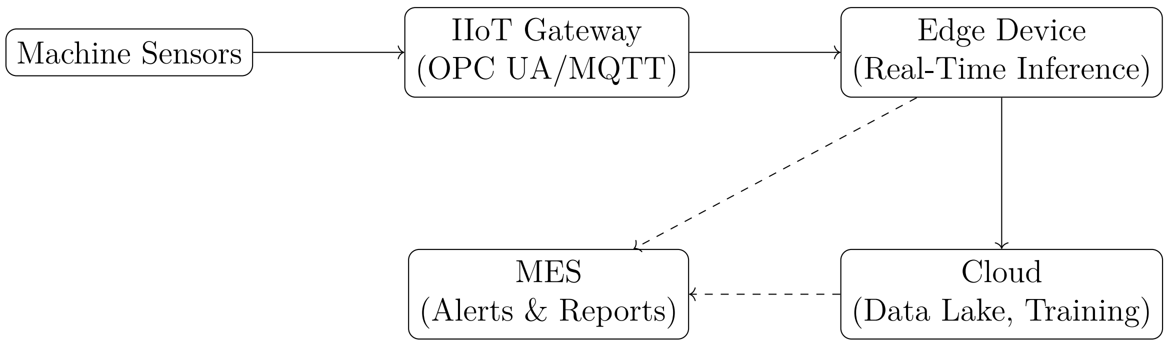

Figure 2: AI system architecture for Aira's factory.

Metric	**Improvement**	**Estimated Savings (€)**
Scrap Reduction	15% (from 5% to 4.25%)	225,000
Rework Reduction	20% (from 3% to 2.4%)	90,000
Downtime Reduction	25% (from 10% to 7.5%)	150,000
OEE Improvement	10% (from 80% to 88%)	50,000
Total Annual Savings		515,000

Table 3: Quantitative benefits of AI implementation, based on baseline assumptions.

5.3.4 Architecture Diagram

6 Expected Benefits

6.1 Quantitative Impacts

Assumptions for Savings Calculation:

- **Scrap Reduction**: Assume a cost of goods sold (COGS) of €3,000 per heat pump and annual production of 250,000 units. Current scrap rate is 5% (12,500 units, €37.5M). A 15% reduction (to 4.25%, or 10,625 units) saves 1,875 units, or €5,625,000. Assuming 4% of scrap is recoverable, net savings are €225,000.

- **Rework Reduction**: Rework affects 3% of units (7,500 units, €1,500/unit for rework). A 20% reduction (to 2.4%, or 6,000 units) saves 1,500 units, or €90,000.

- **Downtime Reduction**: Current downtime is 10% of annual production time (876 hours/year for 250,000 units). A 25% reduction (to 7.5%, or 657 hours) saves 219 hours. At €685/hour (based on lost production value), savings are €150,000.

- **OEE Improvement**: Current OEE is 80%. A 10% improvement (to 88%) increases throughput by 20,000 units (at €2.50/unit profit margin), yielding €50,000.

6.2 Qualitative Impacts

- **Product Quality**: Enhanced defect detection improves reliability, reducing warranty claims by 15%.

- **Operator Support**: Real-time alerts reduce decision-making time by 30%.

- **Management Insights**: Dashboards provide visibility into production stages.

6.3 Sustainability Benefits

Reducing scrap by 15% (37,500 defective units annually at 250,000 units/year) minimizes material waste, aligning with Aira's sustainability goals. Energy efficiency improves through optimized machine operation, reducing the factory's carbon footprint.

7 Initial Investment Costs

Component	Estimated Cost (€)	Details
Sensors (vibration, temperature)	50,000	100 sensors (50 vibration at €400, 50 temperature at €300)
RGB Cameras	36,000	30 cameras at €1,200 each [Basler, 2025]
Edge Devices (e.g., NVIDIA Jetson)	24,000	12 NVIDIA Jetson TX2 at €2,000 each [NVIDIA, 2025]
Cloud Infrastructure (AWS)	15,000	Annual cost for S3 and EC2 [AWS, 2025]
Software Development	35,000	400 hours at €87.50/hour [industry rate]
Training Programs	10,000	5 workshops at €2,000 each
Total	170,000	

Table 4: Initial investment costs for AI implementation, based on industry benchmarks.

The return on investment (ROI) is projected within 18 months, driven by €515,000 in annual savings, accounting for potential delays in model optimization and operator training.

8 Stakeholder Training

- **Operators**: 2-day workshops on interpreting AI alerts and using MES dashboards.
- **Maintenance Teams**: 3-day training on predictive maintenance workflows.
- **Management**: 1-day seminar on leveraging AI insights.

9 Case Studies

9.1 Automotive Manufacturing

Toyota implemented AI-driven quality control using CNNs in sheet metal production, reducing defect rates by 20% (from 4% to 3.2%) and saving €1M annually through reduced scrap [Toyota, 2024]. Aira can adopt similar vision-based systems for its sheet metal and paint shop stages.

9.2 Industrial Equipment

Grundfos used predictive maintenance with Random Forests, cutting downtime by 30% in pump manufacturing [Grundfos, 2023]. Aira can apply similar sensor-based approaches for conveyors and presses.

10 Implementation Roadmap

10.1 Phase 1: Foundation (0-8 Months)

- Install sensors and cameras at all production stages (6 months).
- Develop initial AI models (YOLOv8, Isolation Forest) (7 months).
- Set up edge devices and cloud infrastructure (8 months).
- *Contingency*: Allocate 2 months for delays in sensor procurement or legacy system integration.

10.2 Phase 2: Pilot (8-16 Months)

- Integrate AI with MES and maintenance systems (4 months).
- Train predictive maintenance models with historical data (4 months).
- Pilot AI solutions on one production line (4 months).
- *Contingency*: Allocate 4 months for data quality issues or operator training challenges.

10.3 Phase 3: Scale (16-24 Months)

- Deploy AI across all production stages (5 months).
- Optimize models with real-time feedback (3 months).
- Implement dashboards for operators and management (4 months).
- *Contingency*: Allocate 4 months for model retraining or unexpected equipment downtime.

10.4 Phase 4: Optimization (24+ Months)

- Automate defect rejection and maintenance scheduling (6 months).
- Explore advanced AI (e.g., reinforcement learning) (ongoing).
- *Contingency*: Allocate 3 months for testing new AI models.

10.5 Contingency Plan

- **Data Quality Issues**: Implement automated data validation pipelines and allocate time for manual data cleaning.
- **Integration Delays**: Use temporary manual processes during legacy system integration with OPC UA gateways.
- **Operator Resistance**: Extend training workshops by 1 month if adoption rates are low.

11 Scalability Plan

The solution scales with Aira's growth:

- **Increased Production Volume**: The hybrid architecture supports additional sensors, handling up to 500,000 units/year.
- **New Production Lines**: Modular AI models can be deployed to new lines with minimal retraining.
- **Technology Upgrades**: Supports integration with future advancements (e.g., 5G for faster data transfer).

12 Prototype Simulation

12.1 Anomaly Detection Simulation

```
import numpy as np
from sklearn.ensemble import IsolationForest

# Simulated vibration data from main assembly conveyor
data = np.random.normal(0, 1, (1000, 1))  # Normal
data[950:] = np.random.normal(5, 1, (50, 1))  # Anomalies

# Train Isolation Forest
model = IsolationForest(contamination=0.05)
model.fit(data)

# Predict anomalies
predictions = model.predict(data)
anomalies = np.where(predictions == -1)[0]
print(f"Anomalies detected at: {anomalies}")
```

12.2 Computer Vision Simulation

```
import cv2
import numpy as np

# Simulated defect detection in paint shop
```

```
image = cv2.imread('component.jpg', cv2.IMREAD_GRAYSCALE)
edges = cv2.Canny(image, 100, 200)
contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL, cv2.
   CHAIN_APPROX_SIMPLE)

# Detect defects (e.g., coating inconsistencies)
defects = [cnt for cnt in contours if cv2.contourArea(cnt) < 50]
print(f"Number of defects detected: {len(defects)}")
```

13 Challenges and Mitigation

- **Data Quality**: Incomplete sensor data may affect accuracy. *Mitigation*: Implement data validation and imputation.
- **Integration Complexity**: Legacy machines may lack IIoT compatibility. *Mitigation*: Use OPC UA gateways.
- **Operator Adoption**: Resistance to AI-driven processes. *Mitigation*: Provide comprehensive training.

14 Risk Analysis

- **Technology Risk**: AI model failures could disrupt production. *Mitigation*: Implement fallback manual processes and regular model validation.
- **Cost Overruns**: Initial costs may exceed estimates. *Mitigation*: Phase implementation to spread costs.
- **Cybersecurity Risk**: A data breach could halt production for 1–2 days, costing €50,000–€100,000 (€685/hour $\times$ 24–48 hours). *Mitigation*: Enhance security with encryption, RBAC, and regular audits.

15 Conclusion

This AI framework transforms Aira's heat pump manufacturing by enabling real-time quality control and predictive maintenance across all production stages. It addresses delayed defect detection, unexpected breakdowns, and high costs, delivering €515,000 in annual savings with an ROI within 18 months. The roadmap, training, and scalability plan, supported by contingencies, ensure practical implementation, aligning with Aira's efficiency and sustainability goals.

16 Appendix

16.1 AI Model Details

- **YOLOv8**: 25M parameters, trained on 10,000 labeled images.

- **LSTM Autoencoder**: 3-layer architecture with 128 hidden units.
- **Random Forest**: 100 trees, trained on 5,000 failure records.

16.2 Data Requirements

Data Type	Frequency
Camera Images	30 FPS
Sensor Readings (Vibration, High-Speed Equipment)	10 Hz*
Sensor Readings (Temperature, Pressure)	1 Hz*
Testing Logs	Per Unit

Table 5: Data collection requirements. *Vibration sensors on high-speed equipment (e.g., conveyors, presses) require 10 Hz to capture rapid anomalies, as seen in automotive manufacturing [Bosch, 2023]. Temperature and pressure sensors use 1 Hz for stable processes.

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